

 7. GOVERNANCE & RELEASE GATE · STPA Hazard Mitigation ·
Bumpless Override · Defensible Release Case (**LOOP-01** → **REL-01**)



1. SENSING

Photons / Voltages to DMA
MIPI CSI-2 · SPI Bus Priority



2. PERCEPTION

ViT Encoders · DINO v2
3D Spatial Affordance Tokens



3. MEMORY

Temporal Belief · $SE(3)$ Trees
JEPA / RSSM World Models



5. PLANNING (System 1.5 · MPU/NPU)

Diffusion Policies · ACT Action Chunking
 H -Step Action Horizons · Jerk Continuity



4. REASONING (System 2 · MPU)

Vision-Language Foundation Models · 3D Goals
Leases with Expiring TTL (t_{expire})

THE PROPOSAL-PERMISSION PRIVILEGE BOUNDARY



6. REFLEX (System 1 · MCU)

1 kHz Real-Time Loop · Control Barrier Functions ($h(x) \geq 0$) · Stopping Distance $d_{\text{stop}} \leq d_{\text{clearance}} \cdot \text{Veto}(u_t)$



THE PHYSICAL WORLD ($w_t \rightarrow w_{t+1}$) · Kinetic Momentum · Joule Heat · Friction · Collision Dynamics

Endogenous Sensory Shift (o_{t+1})